

Tiling Spaces of Rational, Nilpotent Lie Groups are Fiber Bundles over Nilmanifolds

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Outline

① Tiling Spaces of $\mathbb{R}^d$

- Simple Tilings

- Tiling Spaces

- Generalizations

② 1-Connected, Nilpotent Lie Groups

- Rational Nilpotent Lie Groups

- Nilmanifolds

- Left-Invariant Metrics

③ Tiling Spaces of Nilpotent Lie Groups

- Simple Tilings Revisited

- The Shape Function Associated to a Tiling Space

- Perturbations of Shape Function

- Rationalizing the Shape Function

- Tiling Spaces are Fiber Bundles

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- 2 Each tile is a polytope.
- 3 If two tiles meet, they meet completely along their cellular faces.

Simple Tilings

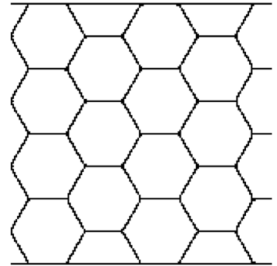
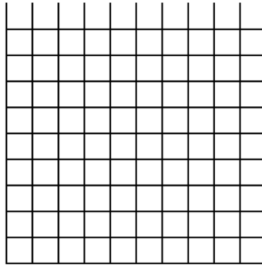
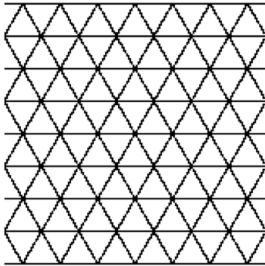


Figure: Periodic Tilings

Simple Tilings

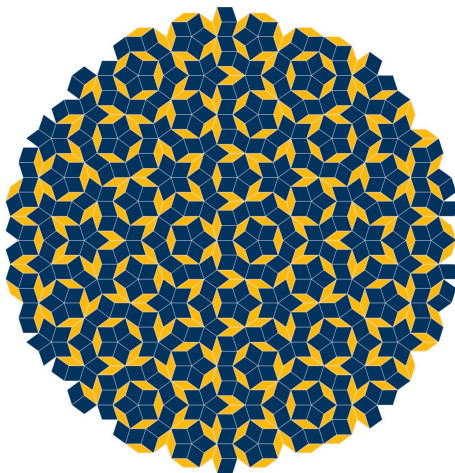


Figure: A Patch of a Penrose Tiling

Violating Hypotheses

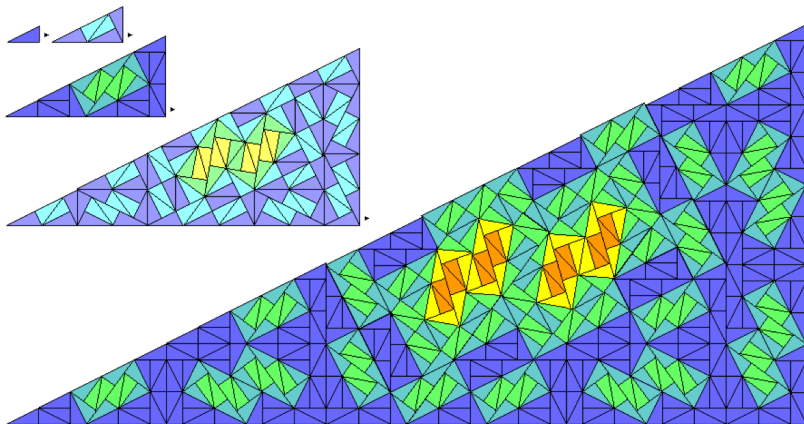


Figure: A Pinwheel Tiling. Lacks finitely many prototiles up to translation.

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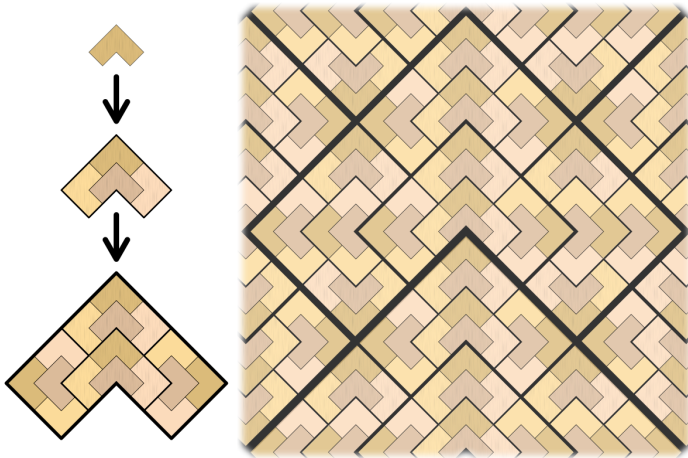


Figure: A chair tiling. Edges don't meet full-face to full-face

Violating Hypotheses

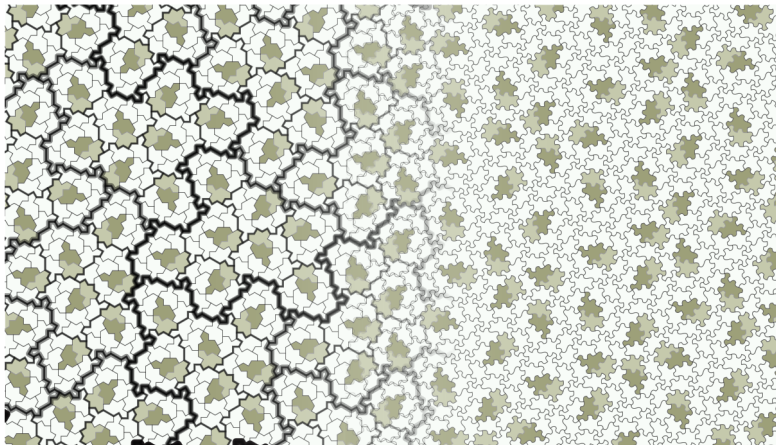
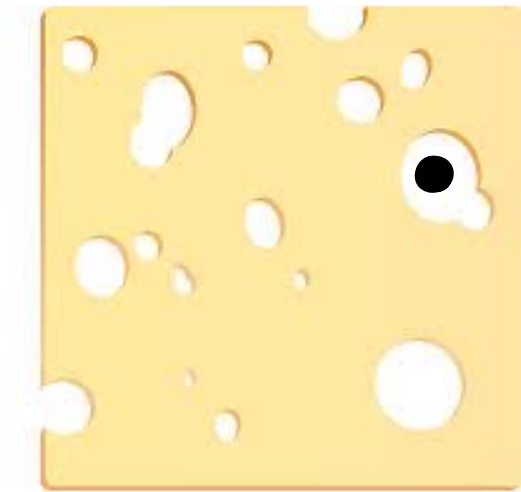
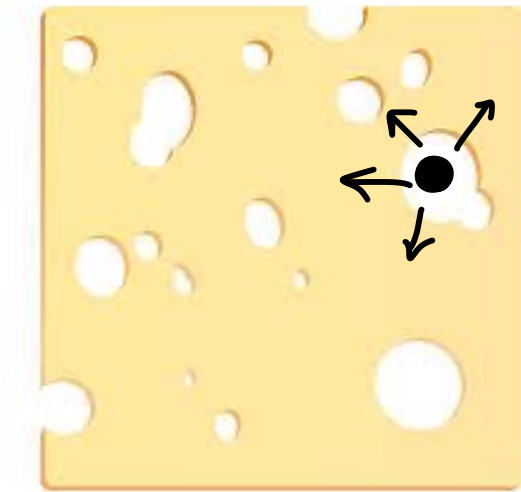


Figure: The "hat" and "spectre" aperiodic tilings. Which is simple?

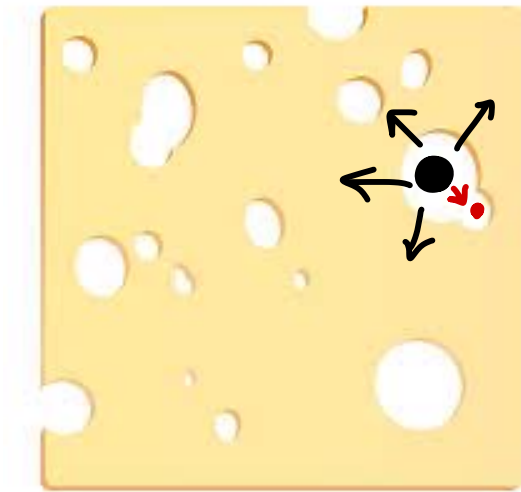
Voronoi Tilings



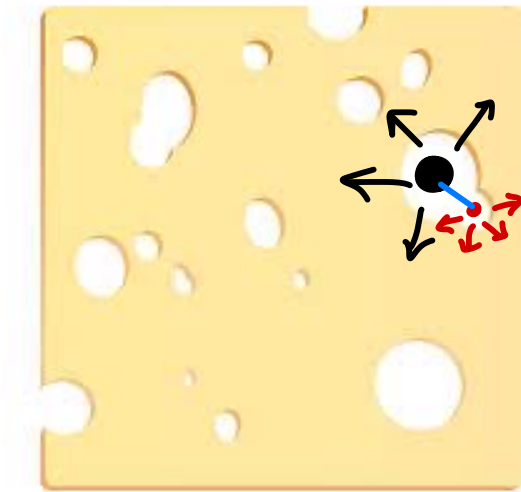
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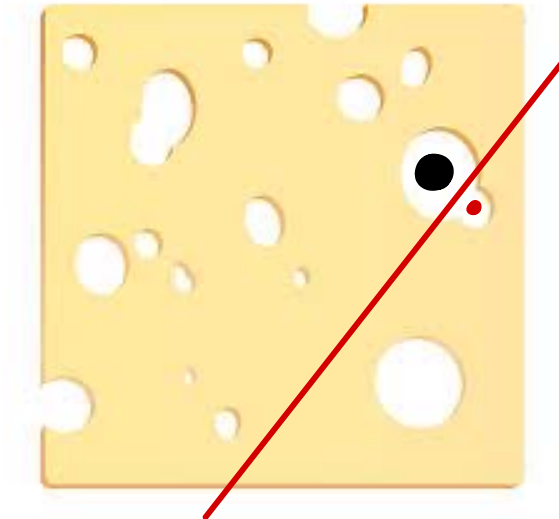
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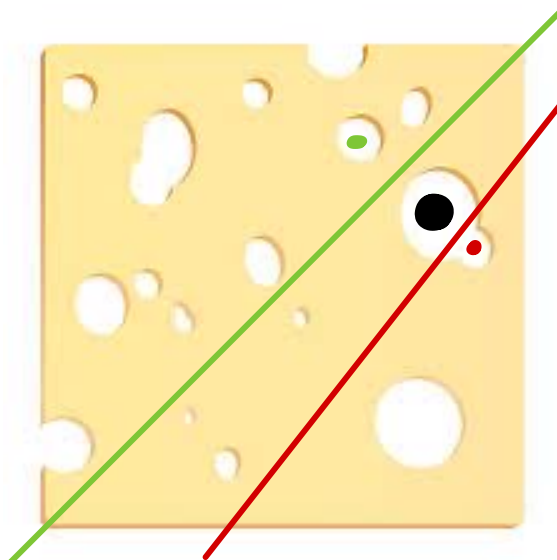
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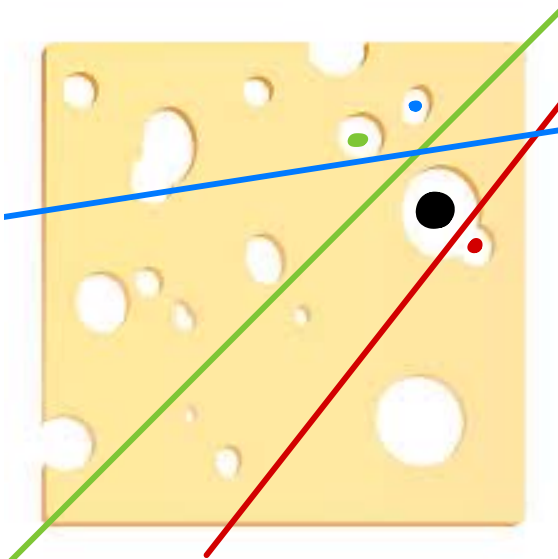
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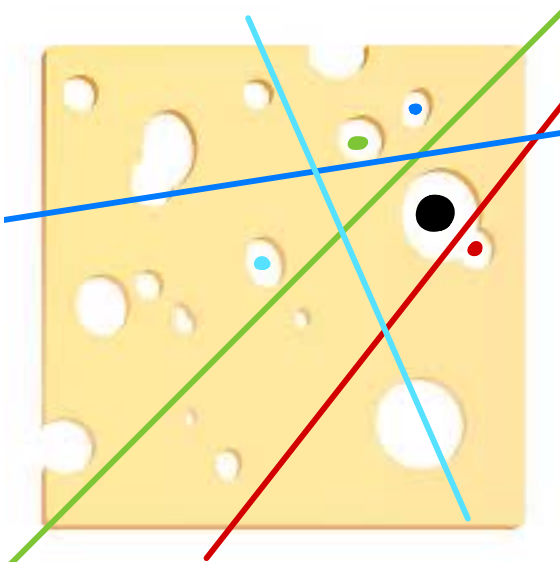
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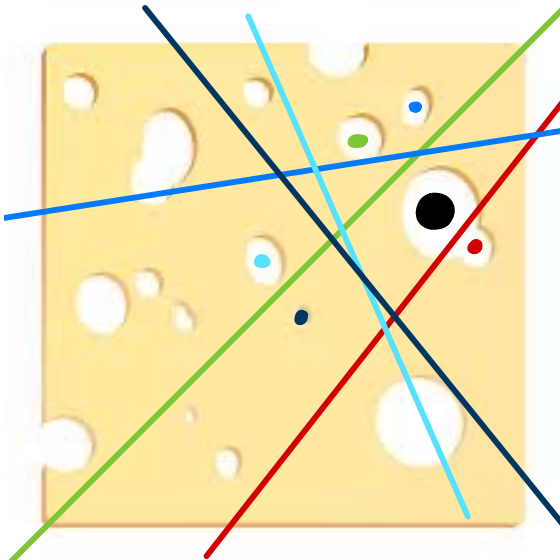
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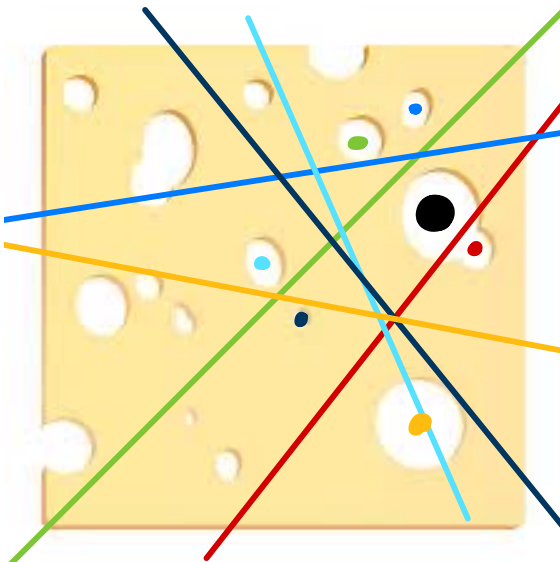
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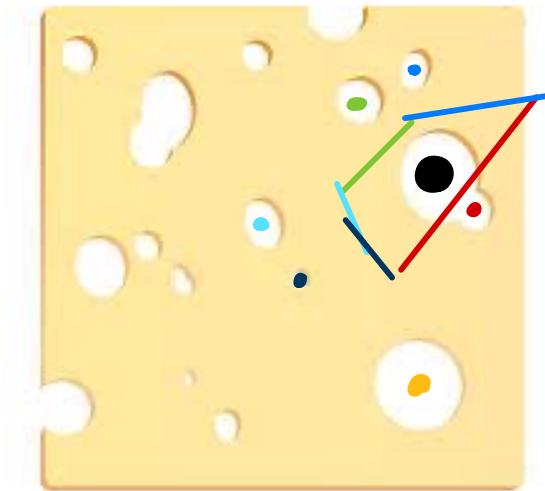
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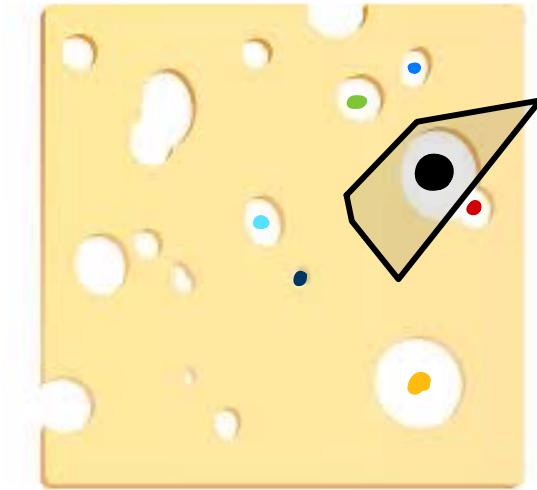
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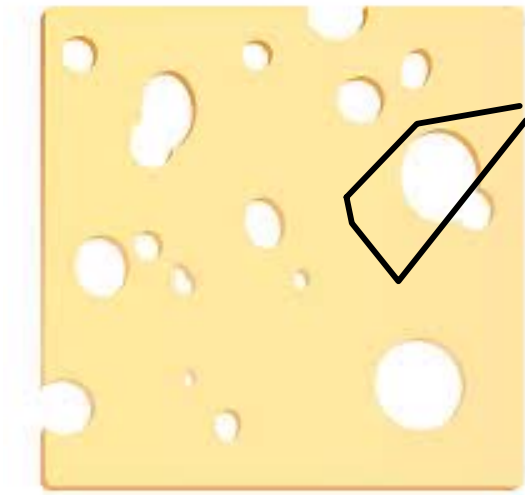
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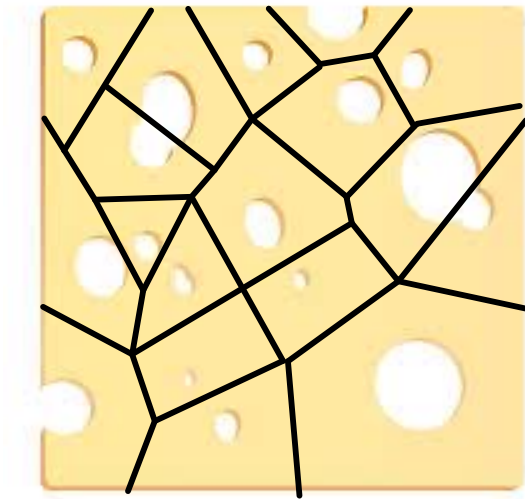
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Voronoi Tilings



Definition (Tiling Metric)

Two tilings $\mathcal{T}$ and $\mathcal{T}'$ of $\mathbb{R}^d$ are ε -close if up to a translation by ε , they agree on a ball of radius ε^{-1} around their basepoints.

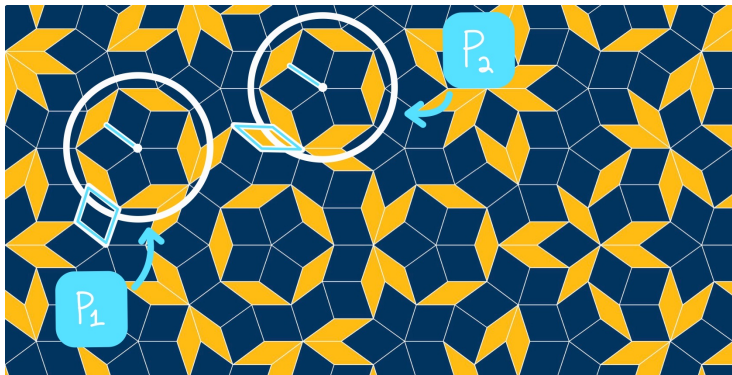


Figure: Patches P_1 and P_2 are about distance $\frac{1}{2}$ apart.

A *tiling space* is a set of tilings which is closed under translation by $\mathbb{R}^d$ and complete in the tiling metric.

Let T be a tiling. The *hull* of T is the set of all those tilings that look like they came from a patch of T at arbitrarily large scales.

FLC Running Assumption

Tilings are assumed to have *finite local complexity*, meaning that there are finitely many prototiles, and finitely many ways tiles are allowed to meet.

Theorem ([Sad03])

Tiling spaces of $\mathbb{R}^d$ are inverse limits of a sequence of compact branched manifolds Γ_n (“Gähler complexes”) and continuous maps $\Gamma_n \rightarrow \Gamma_{n-1}$.

Theorem ([SW03])

Tiling spaces of $\mathbb{R}^d$ are fiber bundles over $\mathbb{T}^d$, typically with Cantor set fiber bundle.

Inverse Limit Structure



Inverse Limit Structure



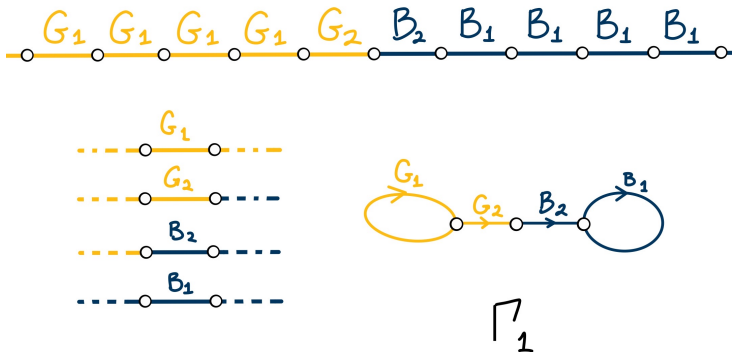
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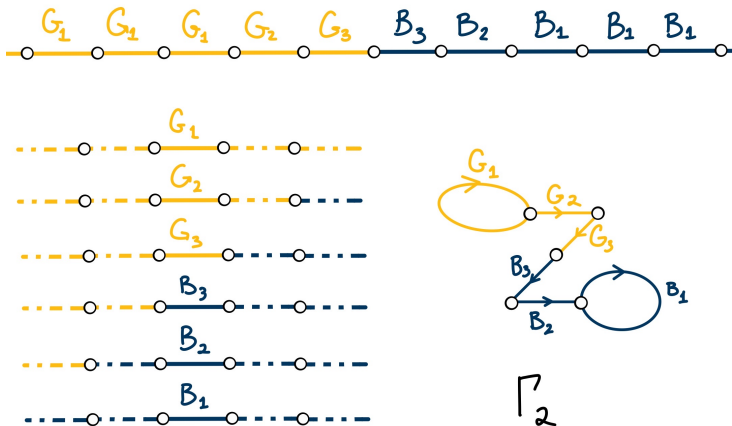
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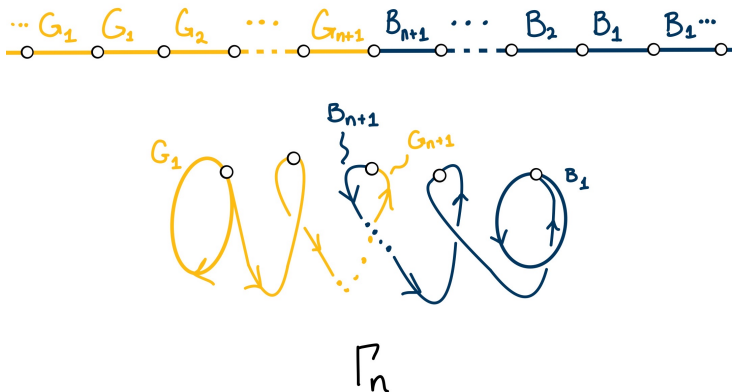
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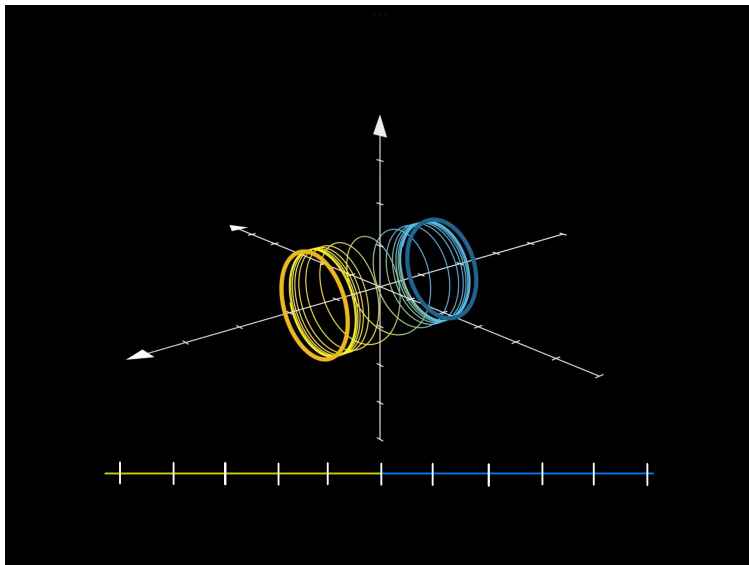


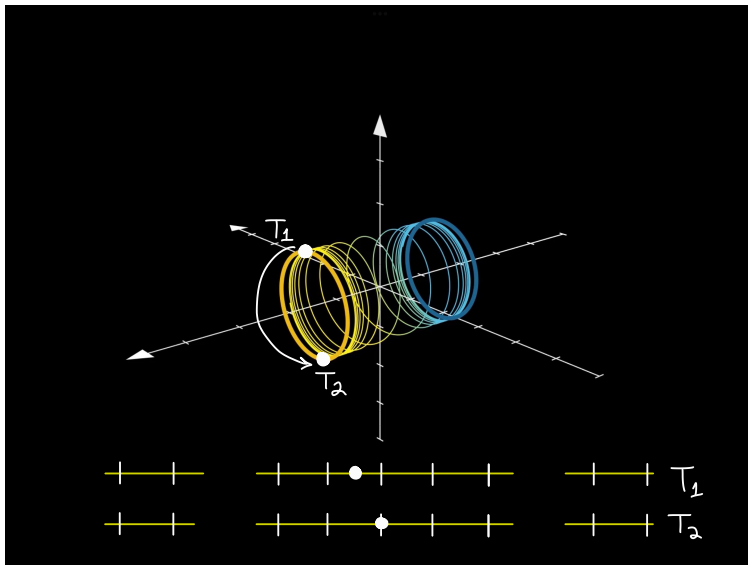
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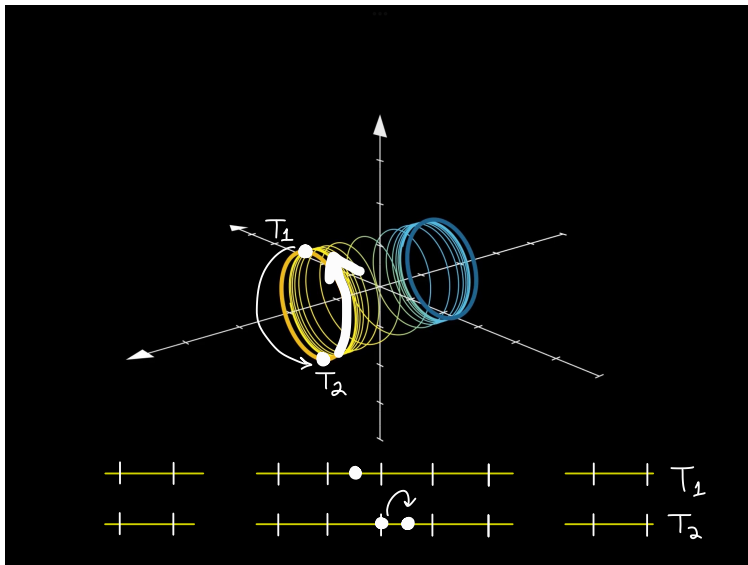


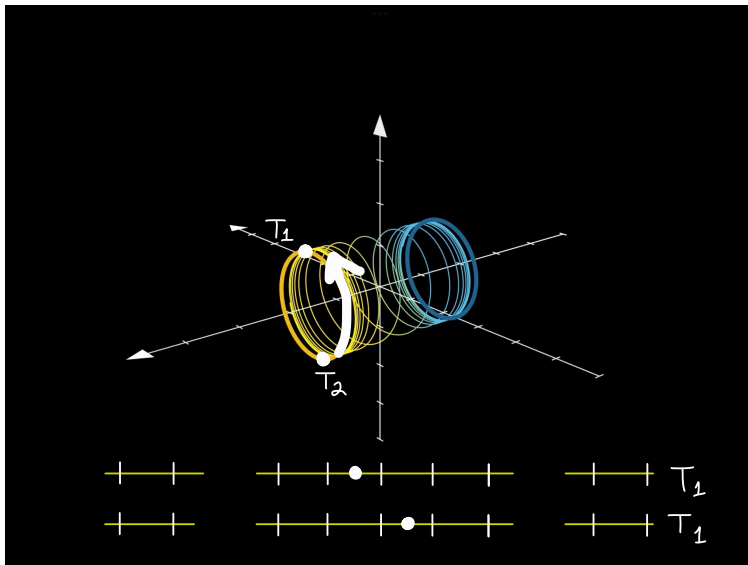
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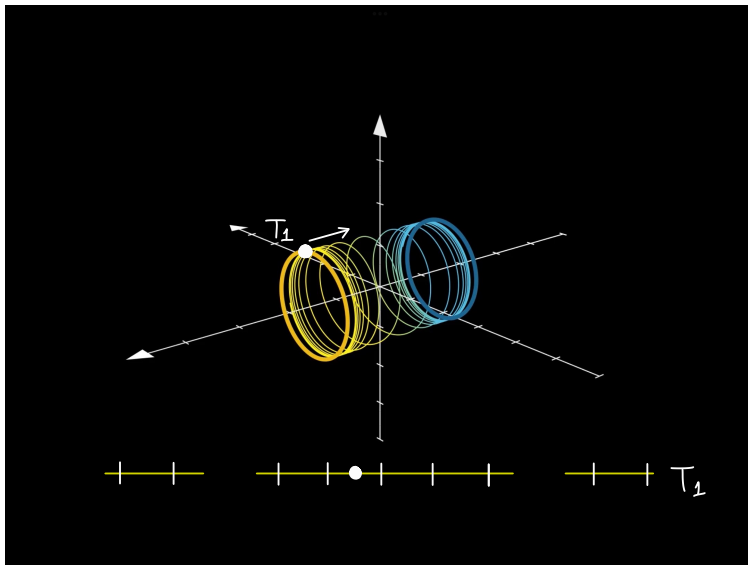


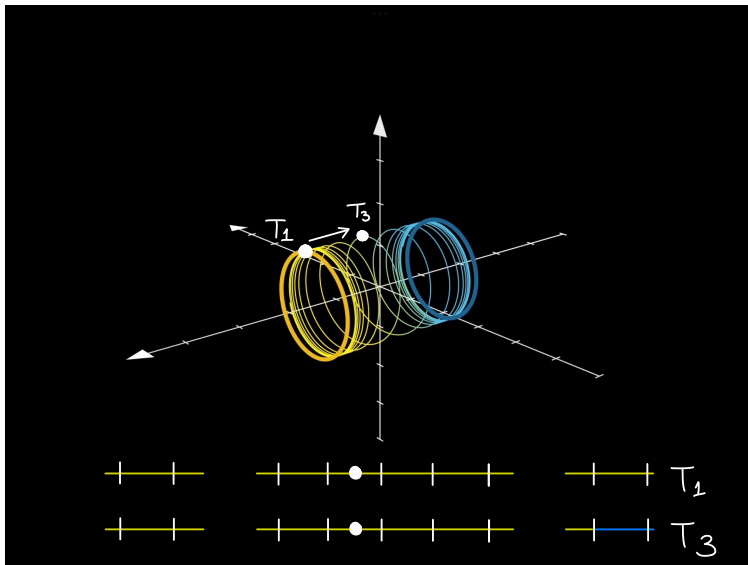


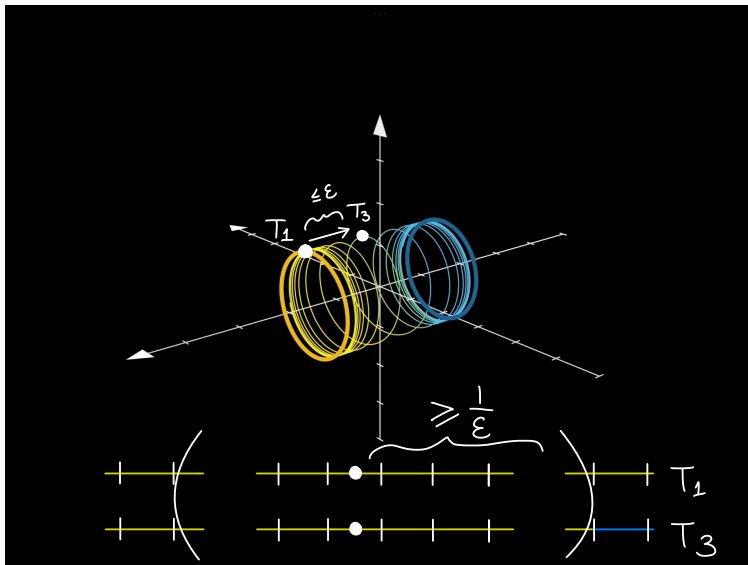












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What if we change $\mathbb{R}^d \curvearrowright \mathbb{R}^d$ to a general $G \curvearrowright M$? What about even just $G = M$ some other Lie group?

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Geometrically Normal Running Assumption

Tilings are still assumed to be FLC, and intersections of tiles are geometrically tame (i.e., smooth topological disks).

The Case for Hope: Past Success

Theorem ([SW03])

If M is a Riemannian homogeneous space and $G \leq \text{Isom}(M)$ is a closed subgroup of isometries, then a G -tiling space over M is the inverse limit of a sequence of branched manifolds $\Gamma_n \rightarrow \Gamma_{n-1}$.

The Case for Hope: Past Success

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Theorem ([Mac18])

Let G be a connected, simply connected, nilpotent Lie group with Lie algebra $\mathfrak{g}$. Then G contains a uniform approximate lattice if and only if $\mathfrak{g}$ has a basis with structure constants in $\overline{\mathbb{Q}} \cap \mathbb{R}$.

In Search of Simplicity: An Outline

Let $G \leq \text{Isom}(M)$ for a Riemannian homogeneous space M .

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Let $G \leq \text{Isom}(M)$ for a Riemannian homogeneous space M .

- 1 Come up with a reasonable definition of a *simple tiling* in this context.
- 2 Explain why geometrically normal tilings of Riemannian manifolds are MLD to simple ones.
- 3 Explain why tiling spaces over certain nilpotent Lie groups (with left-invariant Riemannian metric) are fiber bundles over nilmanifolds.

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Nilpotent Lie Algebras and Lie Groups

A Lie algebra $\mathfrak{g}$ is *nilpotent* if the lower central series terminates:

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supseteq \cdots \supseteq [\mathfrak{g}, [\mathfrak{g}, [\cdots, [\mathfrak{g}, \mathfrak{g}] \cdots]]] = 0.$$

Theorem

A (connected) Lie group $\mathbb{G}$ is nilpotent (as a group in the usual sense) if and only if $\mathfrak{g} := \mathfrak{lie}(\mathbb{G})$ is nilpotent. If $\mathbb{G}$ is simply connected then $\exp : \mathfrak{g} \rightarrow \mathbb{G}$ is a diffeomorphism.

Heuristic: A simply connected nilpotent Lie group $\mathbb{G}^d$ is simply $\mathbb{R}^d$ with an almost-abelian group structure.

The Heisenberg Group

Example

The Heisenberg group $\mathbb{H}$ is the real matrix Lie group

$$\mathbb{H} = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \middle| x, y, z \in \mathbb{R} \right\}.$$

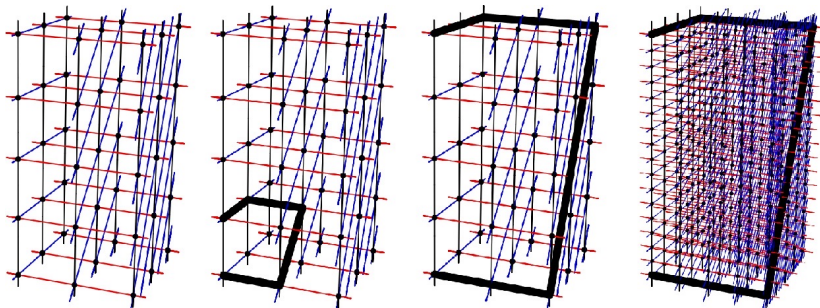
This is diffeomorphic to $\mathbb{R}^3$ with group law

$$(x, y, z) \star (x', y', z') := (x + x', y + y', z + xy' + z').$$

The Lie algebra $\mathfrak{h} = \mathfrak{lie}(\mathbb{H})$ is generated by X, Y, Z with nontrivial bracket

$$[X, Y] = Z.$$

The Cayley Graph of $\mathbb{H}(\mathbb{Z})$



Structure Constants and Rational Lie Groups

Definition

Let $\mathfrak{g}$ be a Lie algebra with basis $\{X_1, X_2, \dots, X_n\}$ and bracket generating relations

$$[X_i, X_j] = \sum_{k < \min\{i, j\}} r_{i,j,k} X_k.$$

The constants $r_k \in \mathbb{R}$ are called the *structure constants* of $\mathfrak{g}$. If $\mathfrak{g}$ has a basis with rational structure constants, we say that $\mathbb{G} := \text{Lie}(\mathfrak{g})$ is *rational*.

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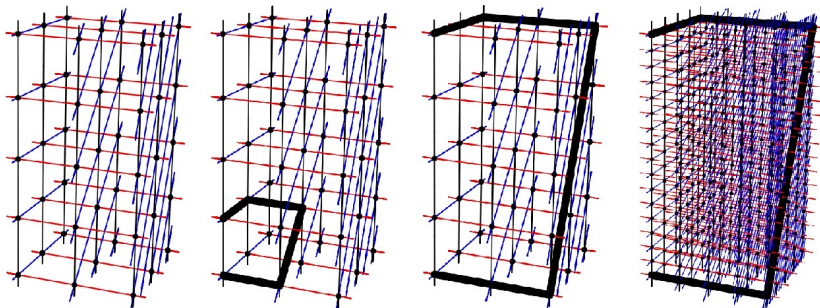
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Theorem ([Mal49])

$\mathbb{G}$ is rational if and only if $\mathbb{G}$ has a uniform lattice. Moreover, exponential coordinates give such a lattice rational coordinates.

The Cayley Graph of $\mathbb{H}(\mathbb{Z})$



Nilmanifolds

Let $\Lambda \subseteq \mathbb{G}$ be a cocompact lattice. Then $\pi_1(\mathbb{G}/\Lambda) = \Lambda$, and $\mathbb{G}/\Lambda$ is called a *nilmanifold*.

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Let $\Lambda = \mathbb{H}(\mathbb{Z})$ be the integer lattice in $\mathbb{G} = \mathbb{H}$. Then $\mathbb{H}/\Lambda$ is called the *Heisenberg Manifold*, and has $\pi_1(\mathbb{H}/\Lambda) = \mathbb{H}(\mathbb{Z})$.

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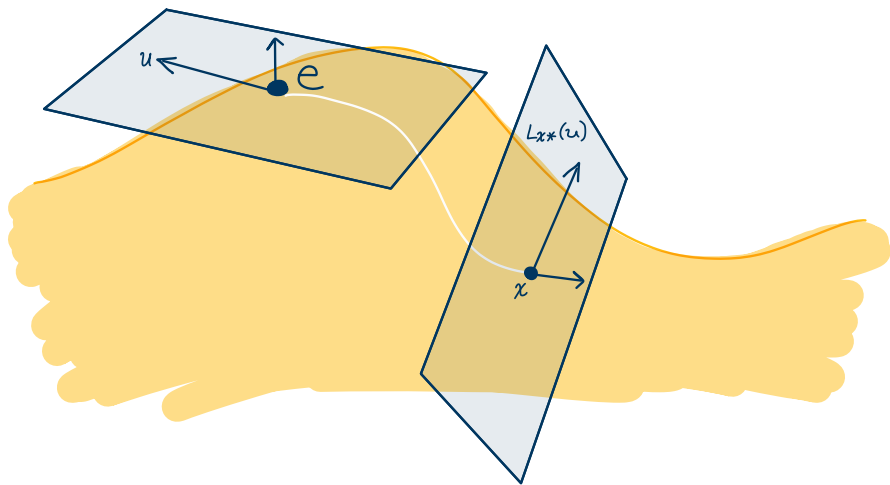
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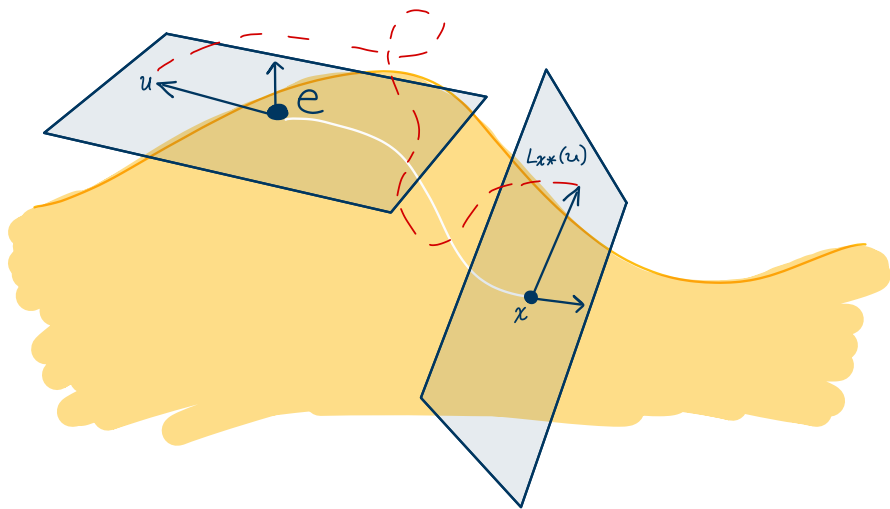
It can be shown that

$$H^k(\mathbb{H}/\Lambda; \mathbb{R}) = \begin{cases} \mathbb{R} & k = 0, 3 \\ \mathbb{R}^2 & k = 1, 2 \end{cases}$$

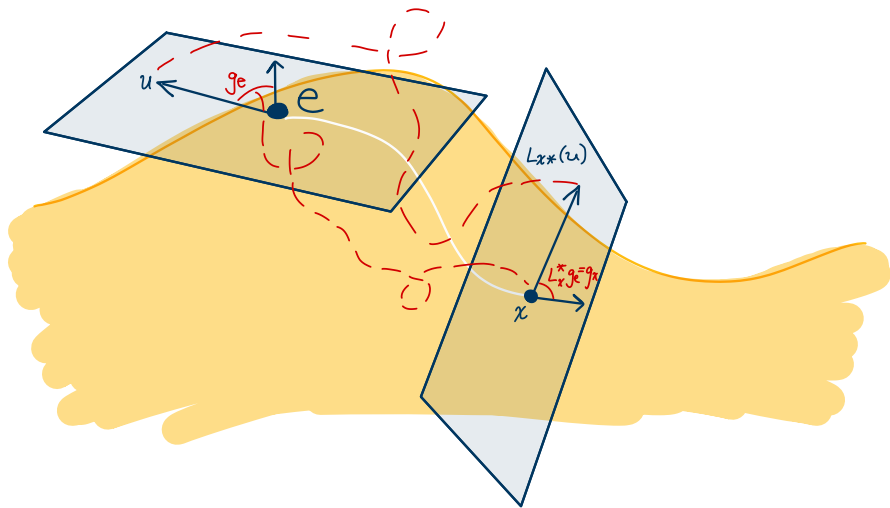
Left-Invariant Riemannian Metrics



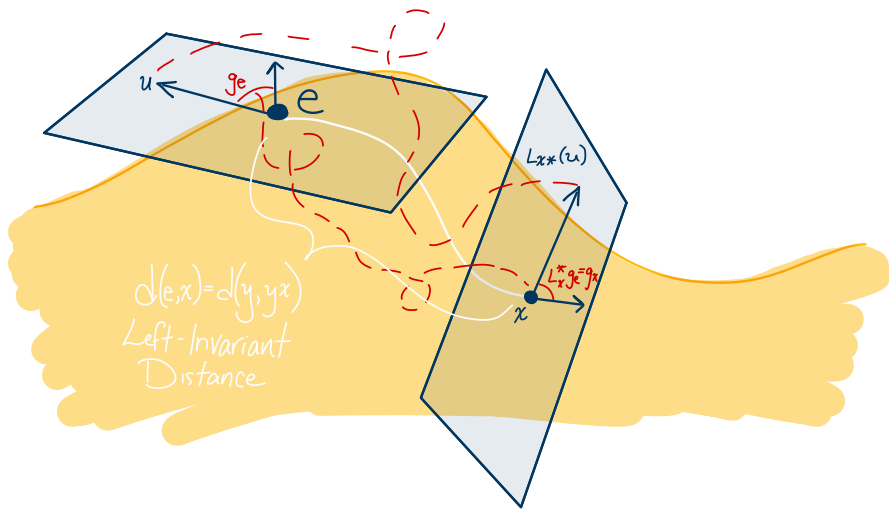
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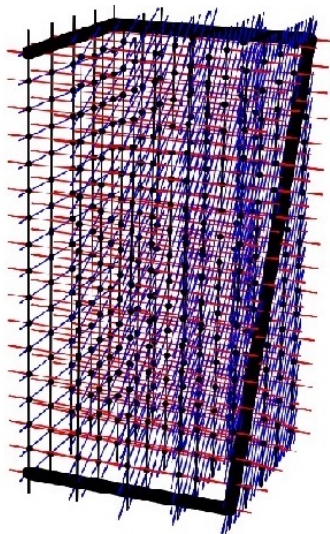
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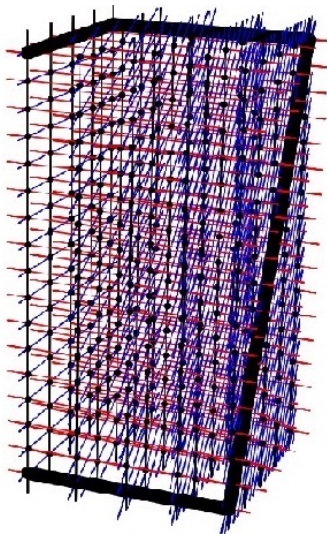


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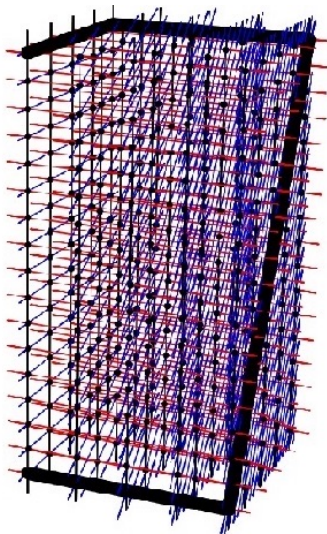


Left-Invariant Riemannian Distance on $\mathbb{H}$



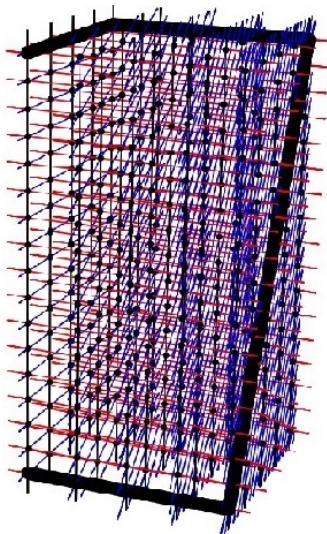
Left-Invariant Riemannian Distance on $\mathbb{H}$ 

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Other (“sub-Riemannian”) metrics allow for dilation that respects the distance. For example, $d_{CC}(0, (0, 0, T^2)) = T$ for all T . This is not the case for Riemannian metrics.

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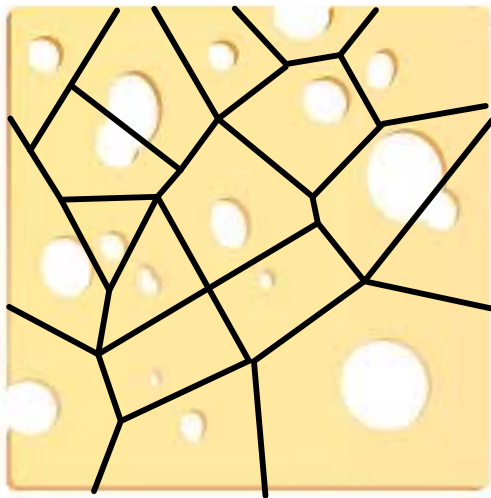
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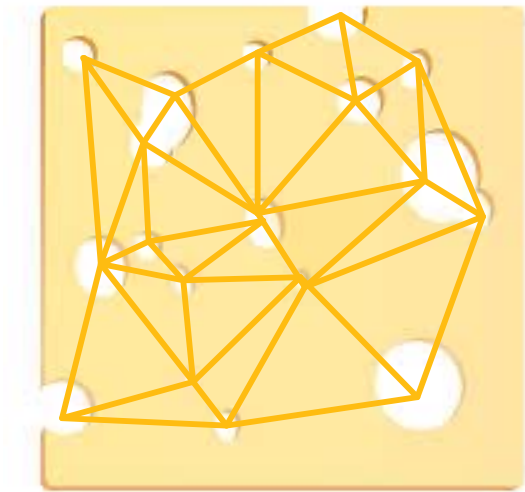
Delaunay Tilings



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Delaunay Tilings



Tilings of $\mathbb{G}$

Let $G \leq \text{Isom}(\mathcal{M})$ for a Riemannian homogeneous space $\mathcal{M}$ (e.g. $G = \mathcal{M} = \mathbb{G} = \mathbb{G}^d$ a connected, simply connected, nilpotent Lie group with left-invariant Riemannian metric)

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- 3 If two tiles meet, they meet completely along their cellular faces.

Simple Tilings aren't Too Restrictive

Heuristically, a Delaunay set D is *generic* if there is an algorithm (e.g. akin to [BDG15]) for producing a triangulation of $\mathcal{M}$ out of local patches of D . This triangulation must be combinatorially invariant under sufficiently small local perturbations of D .

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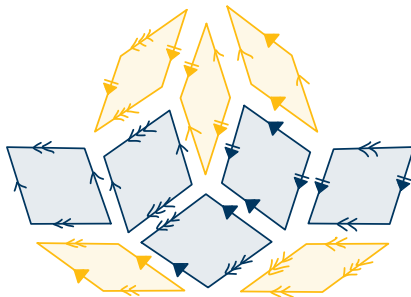
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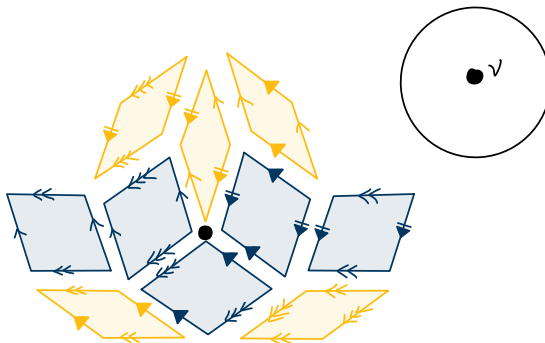
Proof.

Inductively extend a generic Delaunay set in a neighborhood of $\Gamma^{(k-1)}$ to a neighborhood of $\Gamma^{(k)}$. The resulting Delaunay set on Γ lifts to a "pattern equivariant" generic Delaunay set on the original tiling of $\mathcal{M}$. □

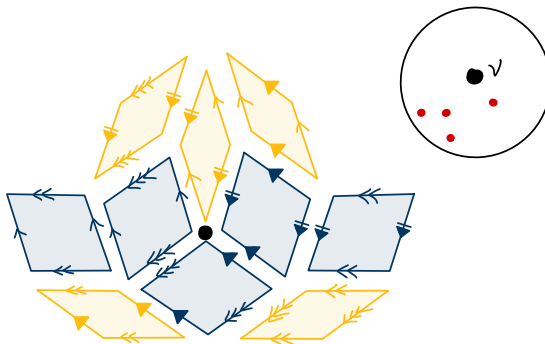
Constructing a Simple Tiling



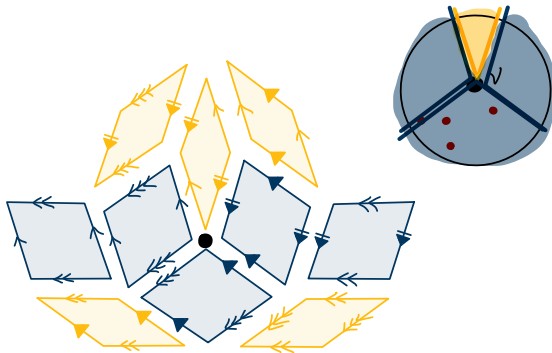
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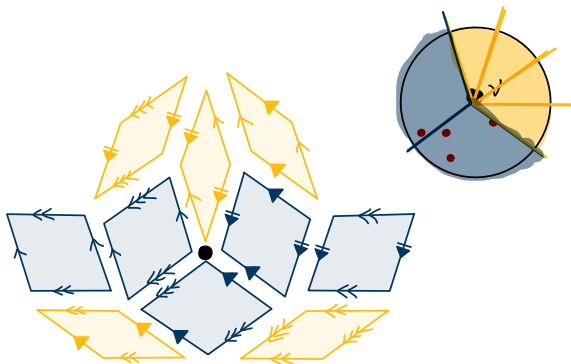
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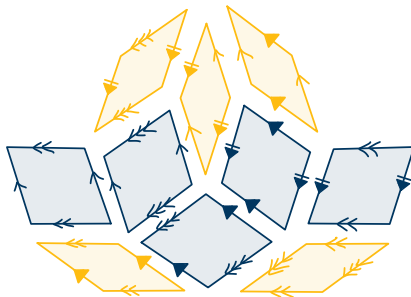
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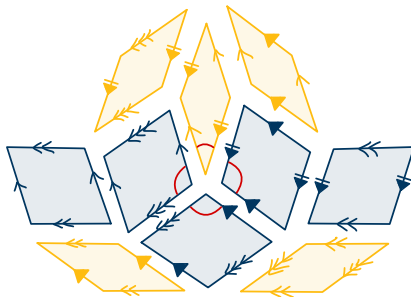
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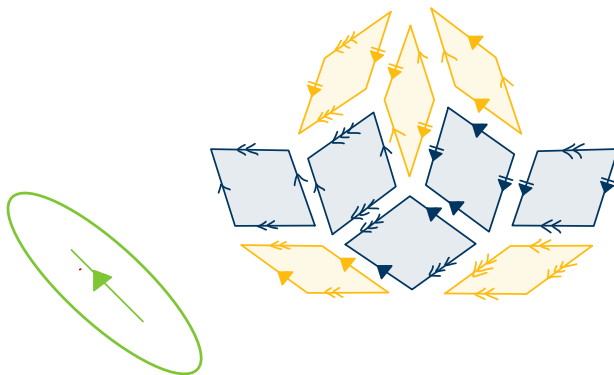
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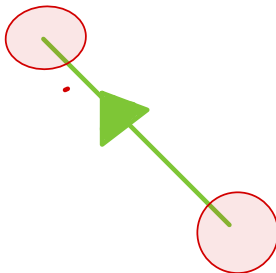
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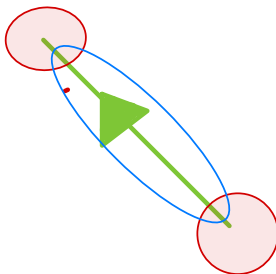
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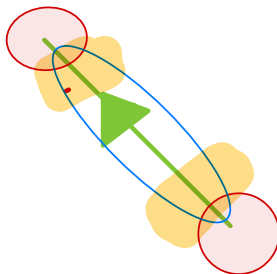
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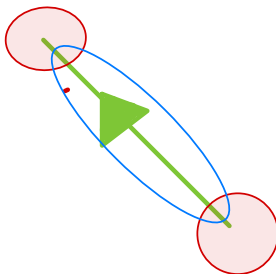
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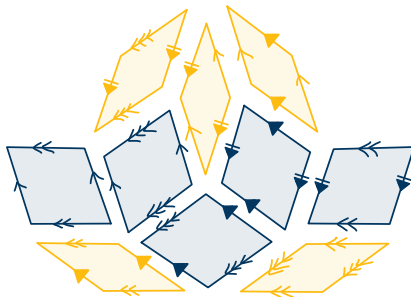
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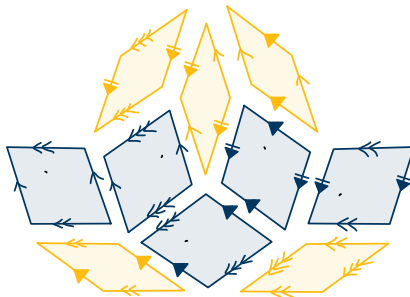
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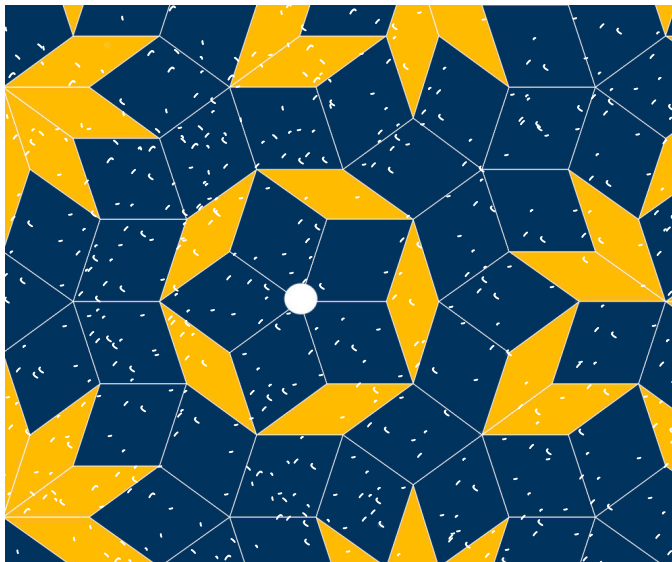
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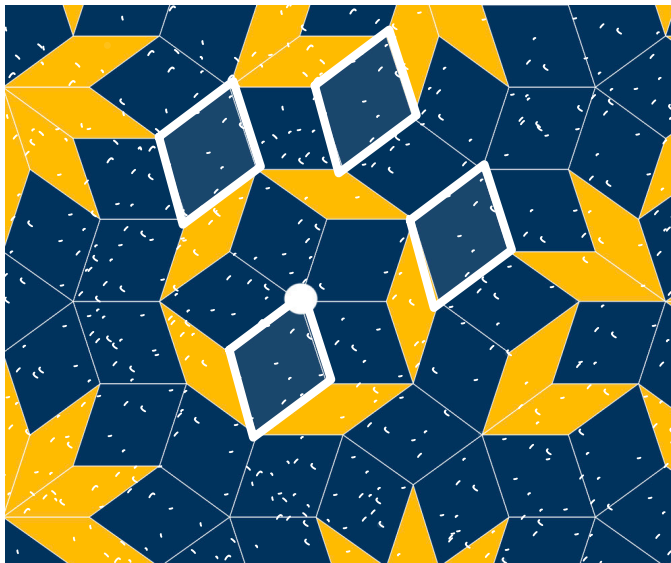
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Shape Functions

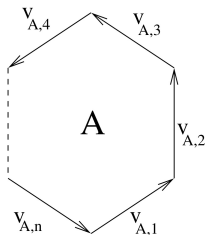


Figure: The vectors $v_{A,i}$ determine the shape of the tile A .

Shape Functions

Let $\mathcal{T}$ be a $\mathbb{G}$ -tiling space with prototiles $\mathcal{P}$ and Gähler complex Γ .

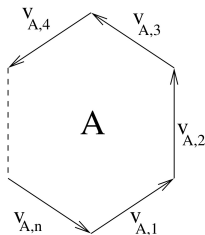
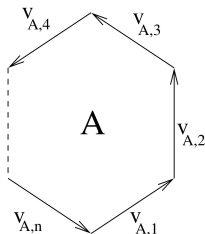


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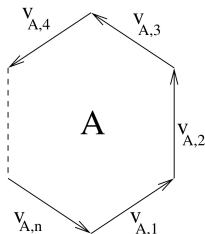


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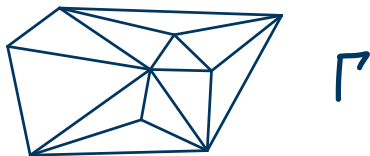
Definition

The *shape function associated to $\mathcal{T}$* is the shape function $s = s_{\mathcal{T}}$ with $s(E) \in \mathbb{G}$ the displacement of any representative of the (oriented) edge E .

Perturbations of Shape Functions

Lemma

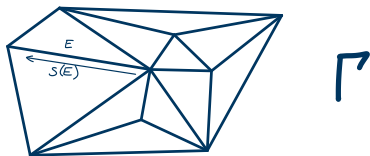
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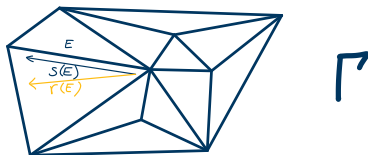
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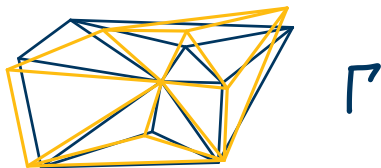
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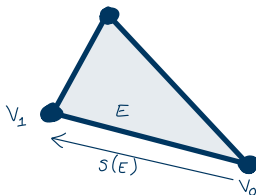
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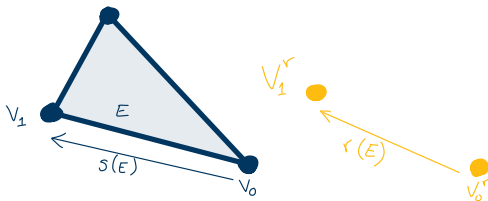
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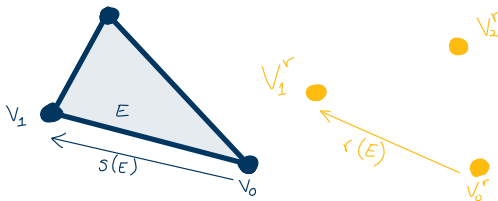
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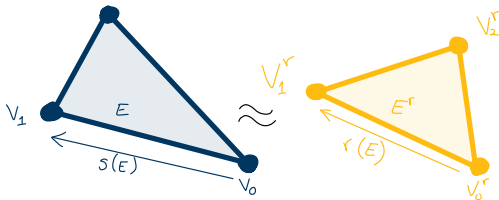
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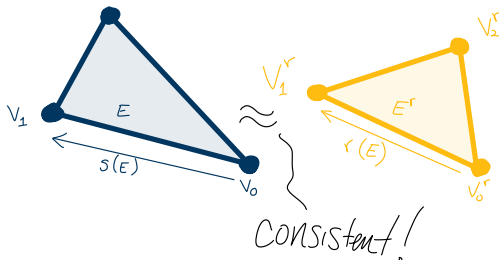
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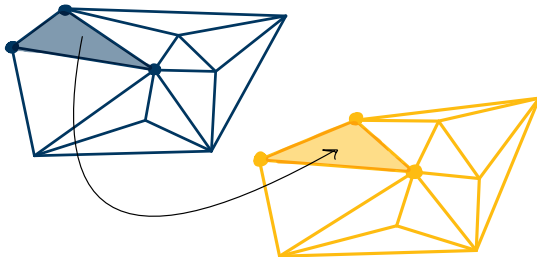
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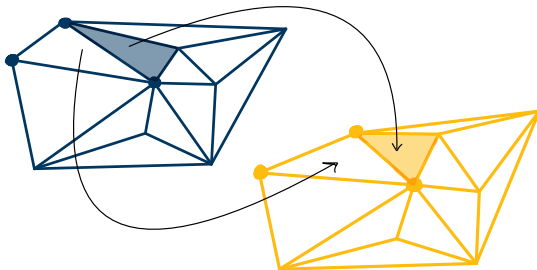
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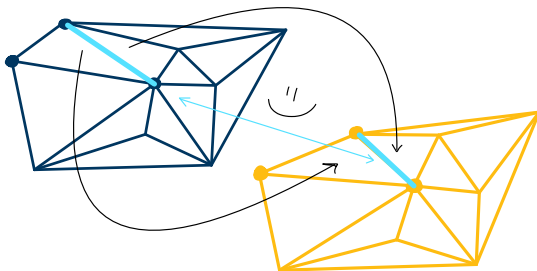
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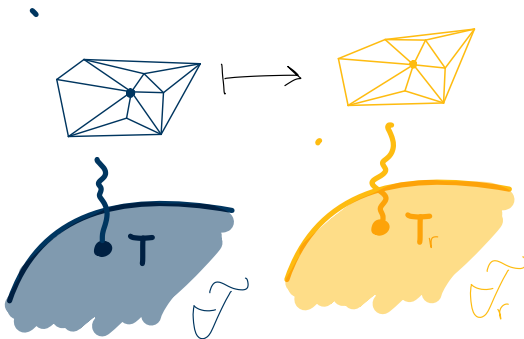
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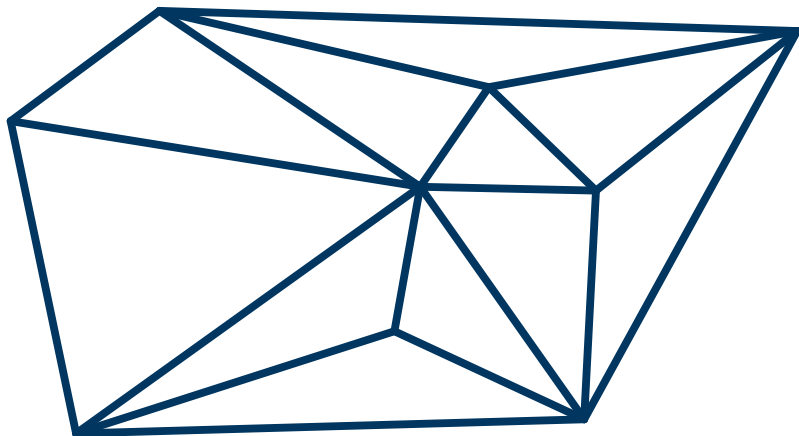
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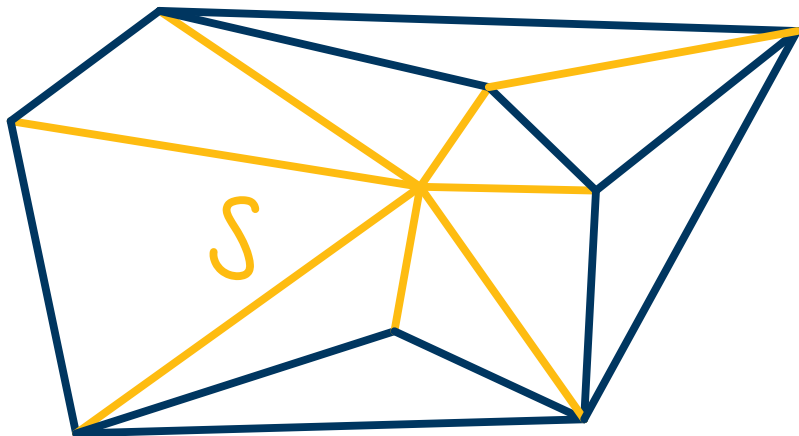
Corollary

Every simple tiling space is homeomorphic to a rational one.

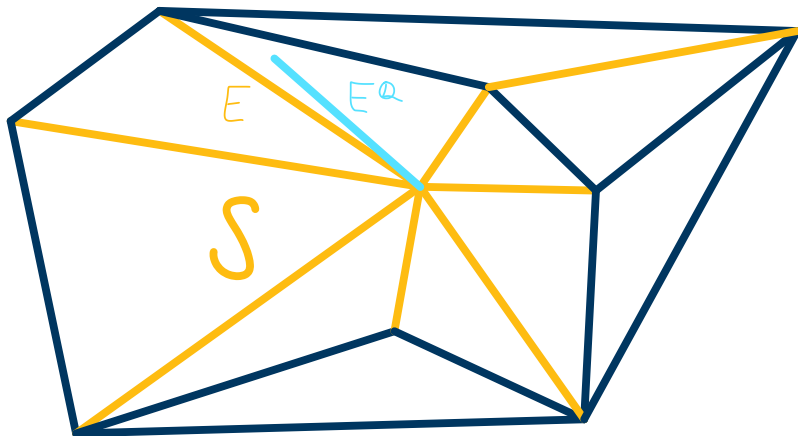
Rationalizing Displacement



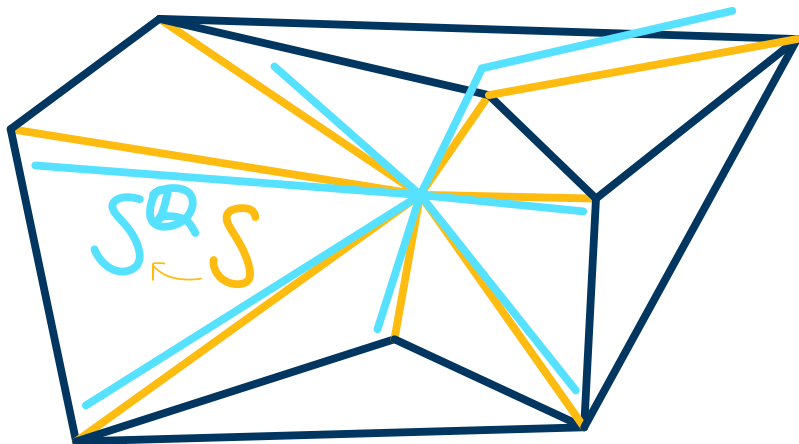
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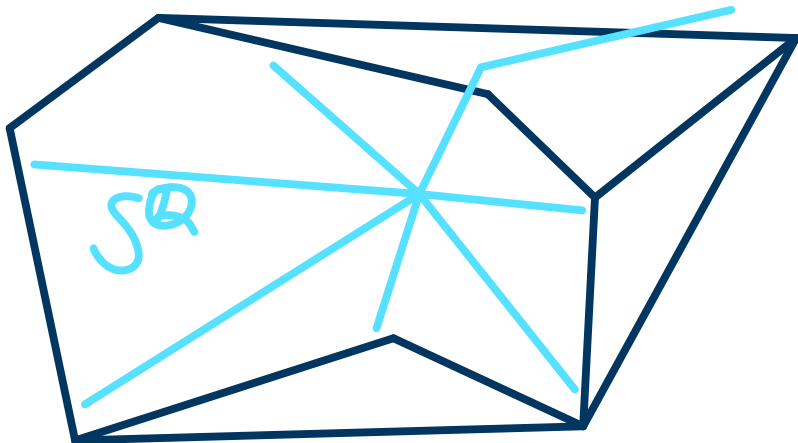
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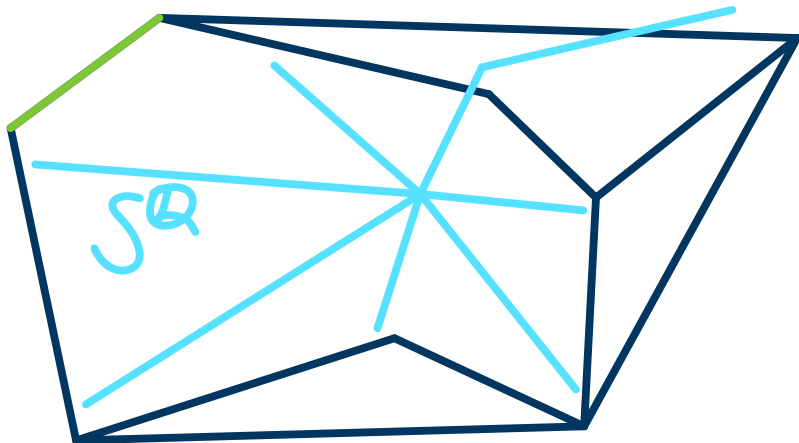
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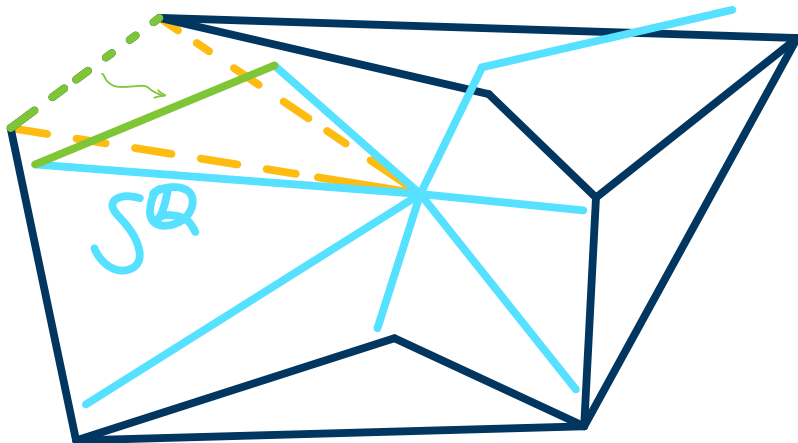
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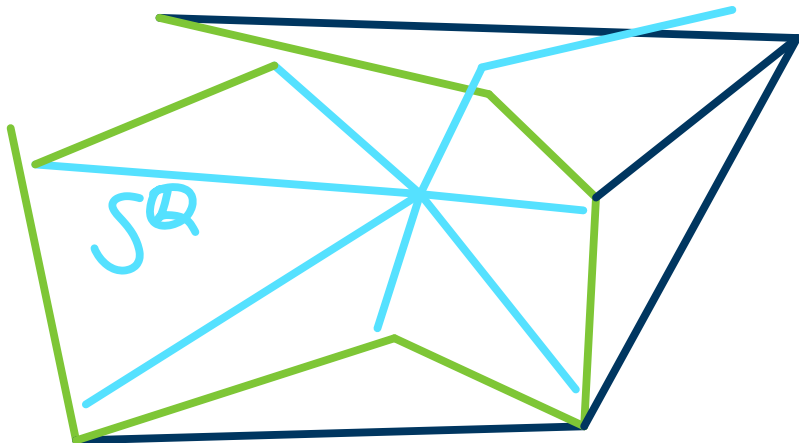
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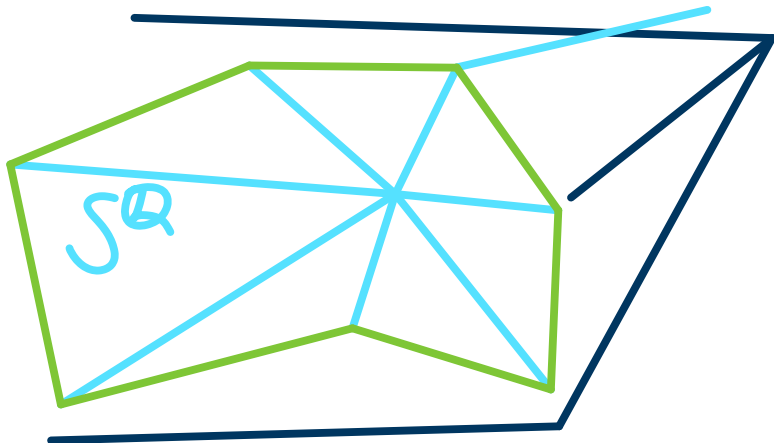
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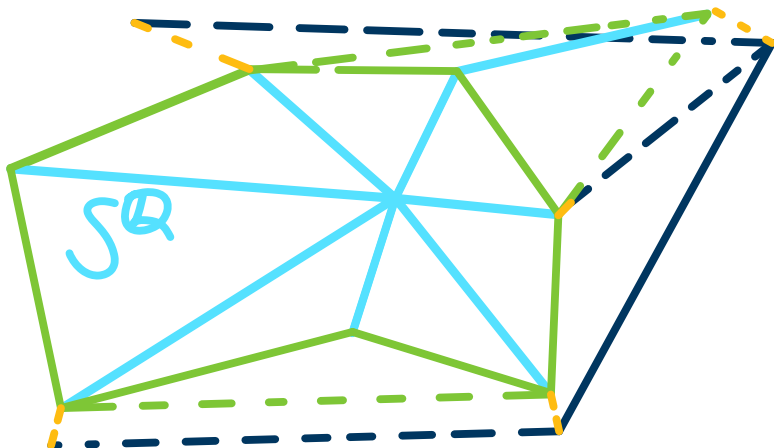
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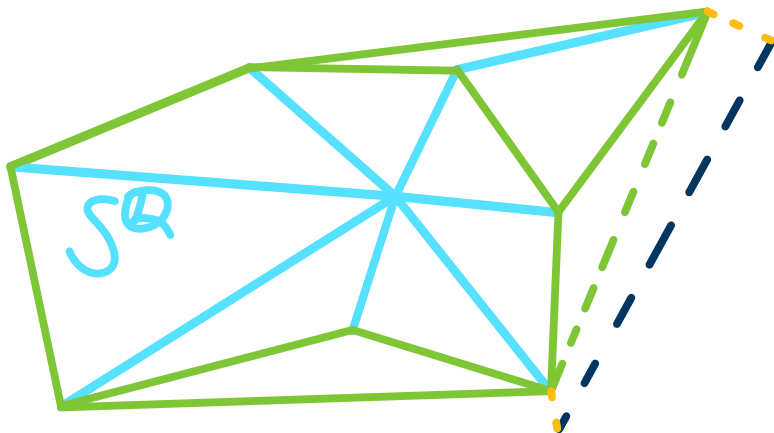
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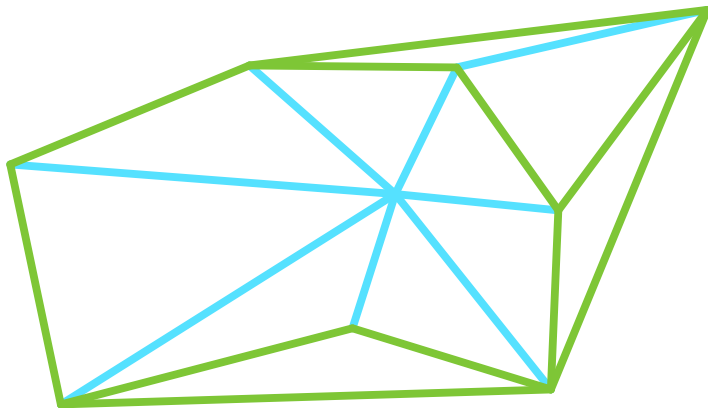
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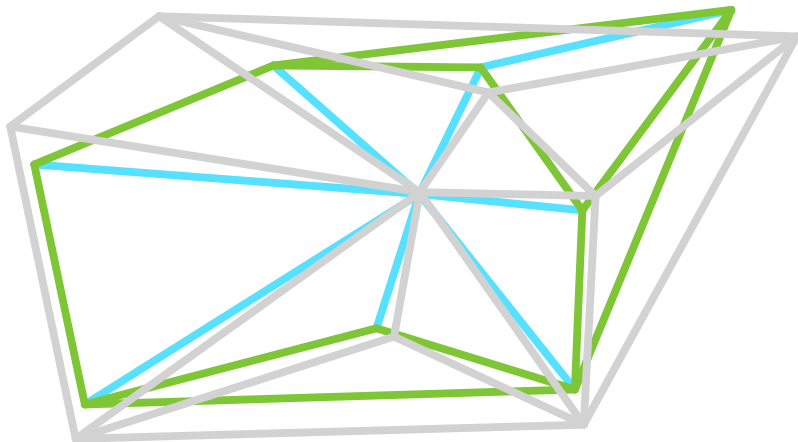
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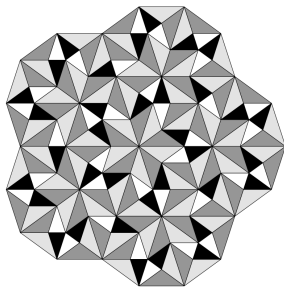
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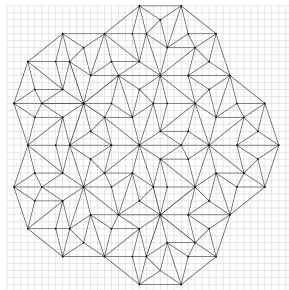
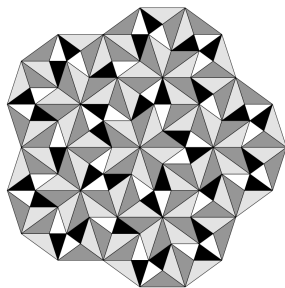
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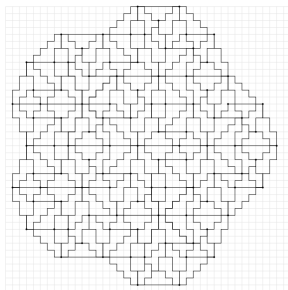
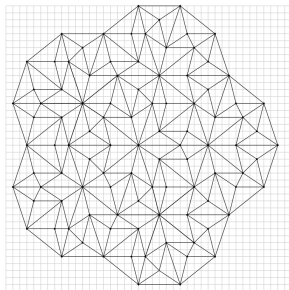
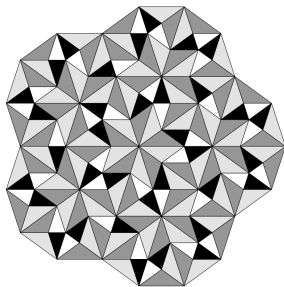
Tiling Spaces are Fiber Bundles



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Theorem

Let $\mathcal{T}$ be a tiling space over a connected, simply connected, rational nilpotent Lie group $\mathbb{G}$. Then $\mathcal{T}$ is fiber bundle over a nilmanifold.

Proof.

It is enough to show this for a rational tiling space $\mathcal{T}$.

Tiling Spaces are Fiber Bundles

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Let $\mathcal{T}$ be a tiling space over a connected, simply connected, rational nilpotent Lie group $\mathbb{G}$. Then $\mathcal{T}$ is fiber bundle over a nilmanifold.

Proof.

It is enough to show this for a rational tiling space $\mathcal{T}$. Let $\mathcal{D}$ be the displacements of $\mathcal{T}$ and let $\Lambda \subseteq \mathbb{G}$ be the lattice generated by $\mathcal{D}$ (see [Cor17]).

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- It can be shown that $H_{PE}^*(\mathcal{T}) \approx \check{H}^*(\Omega_{\mathcal{T}}; \mathbb{R})$ for simple tilings. We expect that this can be used in the study of the topology of tilings of nilpotent spaces, as has been done in the Euclidean setting.

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Thank you!